TASK 2 — Methodology Passport Card

The Impact of Social Media Advertising on University Students' Online Shopping Behavior

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Year: 2026

GitHub: https://github.com/shabdanbekmeruert-commits/social-media-shopping-research

1. Research type and purpose

We chose a quantitative research approach because we want to work with answers that can be measured and compared. We will ask university students about the advertisements they see on social media and their online shopping habits. Most questions will use a 1–5 response scale, while a few questions will ask about actual purchase experiences.

Our study will be cross-sectional: each student will complete the questionnaire once. We are not assigning students to different advertising conditions, so this is an observational survey rather than a controlled experiment. We can examine relationships between variables, but we cannot prove that advertising directly causes a purchase.

Our main goal is to find out whether advertising exposure and perceived credibility are related to students' purchase intention and reported online shopping decisions. We also want to understand whether a positive attitude towards advertisements is linked to more engagement with them.

2. Research questions and hypotheses

We kept the three main research questions from Task 1. For Task 2, we describe each hypothesis in a way that can be checked using survey results.

RQ1. Perception and engagement

Research question: How do university students perceive and interact with social media advertising when shopping online?

H0: There is no association between students' perception of social media advertising and their engagement with advertisements ($\rho = 0$).

H1: A more positive perception of social media advertising is associated with higher advertisement engagement ($\rho > 0$).

RQ2. Advertising exposure and purchase intention

Research question: How does exposure to social media advertising relate to students' online purchase intentions?

H0: There is no association between advertising exposure and online purchase intention ($\rho = 0$).

H1: Higher advertising exposure is associated with higher online purchase intention ($\rho > 0$).

RQ3. Credibility and purchase decisions

Research question: How does perceived credibility of social media advertising relate to students' online purchase decisions?

H0: Perceived advertising credibility is not associated with the probability of a reported online purchase ($\beta = 0$).

H1: Higher perceived advertising credibility is associated with a higher probability of a reported online purchase ($\beta > 0$).

Here, ρ means Spearman's rank correlation coefficient, and β means the coefficient for credibility in a logistic regression model. We will test the hypotheses using a significance level of 0.05, while also reporting effect sizes and sample sizes.

3. Variable matrix

Independent variables describe advertising-related factors. Dependent variables describe students' responses or shopping behavior. Control variables help us check whether other characteristics could explain part of the observed relationship. These are planned measurements; we will finalize the questionnaire before collecting real responses.

Role	Variable	Planned measurement
Independent	Advertising perception	Mean of 1–5 agreement items about how useful, relevant or interesting ads seem (RQ1).
Independent	Advertising exposure	1–5 frequency score, from rarely to very often seeing social media ads (RQ2).
Independent	Perceived credibility	Mean of 1–5 items about how believable and trustworthy ads appear (RQ3).
Dependent	Advertisement engagement	1–5 frequency of clicking, saving, sharing or checking products after seeing ads (RQ1).
Dependent	Purchase intention	1–5 willingness to buy a product online after seeing a social media ad (RQ2).
Dependent	Reported purchase decision	0 = no, 1 = yes: purchased an advertised product online within a defined recent period (RQ3).
Control	Social media use	Approximate daily time spent on social media; collected in categories.
Control	Online shopping frequency	Number or category of online purchases within a defined recent period.
Control	Budget / income	Optional broad spending-budget category; respondents may choose not to answer.
Control	Previous brand experience	Whether the respondent had purchased from or knew the advertised brand before.

4. Survey design and sample

We plan to use a short online questionnaire, for example in Google Forms. Participants will be university students who use social media. Participation will be voluntary, and we will use convenience sampling because it is practical for a student research project. This means the findings may not represent all university students.

Before sharing the form widely, we will ask a few volunteers to check whether the questions are easy to understand. We will use the same question wording and scale directions for everyone. If a variable is measured by several questions, we will calculate a mean score only when enough items are answered.

We will record the number of invited students if it is available, the number of submitted responses and the number of usable responses. We will not present a target sample size as an achieved result. The final number of participants will be reported after data collection.

5. Primary metric, secondary metrics and baseline

Primary metric

Our primary metric is Spearman's rank correlation coefficient (ρ) between advertising exposure and purchase intention (RQ2). We chose it because both variables are planned as ordered 1–5 scores. A positive value means that students who report seeing more advertisements also tend to report stronger purchase intention. We will report ρ, the p-value and the number of valid responses.

Secondary metrics

For RQ1, we will calculate Spearman's correlation between advertising perception and engagement. For RQ3, we will compare the reported purchase rate across credibility scores and, if the data are sufficient, fit a logistic regression model. We will also show response counts, percentages and summary statistics to make the findings easier to interpret.

Baseline

For the main correlation test, our statistical baseline is no monotonic association ($\rho = 0$). For the optional purchase-decision prediction, the baseline will be an intercept-only logistic model that predicts from the overall purchase rate without using advertising variables. We will compare any extended model against this simple baseline using a clearly stated metric, such as log loss. We will not claim that a better prediction model proves causality.

6. Guardrail metrics and quality checks

We will check data quality before calculating the main results. These checks are important because even a correctly written Python script can give misleading results when the input data contain mistakes.

Check	Rule / target	What we will do
Missing required values	Target: no more than 5% missing for key fields	Report the rate; exclude missing pairs from each correlation and show the valid sample size.
Valid ranges	Scale answers must be 1–5; purchase decision must be 0 or 1	Flag invalid values instead of silently treating them as real answers.
Duplicate IDs	0 duplicate response IDs	Flag duplicate IDs for review before analysis.
Minimum analysis size	At least 5 complete rows for a pipeline smoke test	Do not interpret the 5–10 synthetic test rows as research evidence.
Privacy	No names, phone numbers or email addresses in the public dataset	Use anonymous IDs and publish only synthetic sample data.

The 5% missing-value limit is our quality target, not a guarantee. If the collected data do not meet it, we will describe the issue honestly and explain how it affects the analysis.

7. Data analysis and reproducibility plan

First, we will export questionnaire answers to CSV format. Next, the Python script will read the file, check required columns, identify missing or invalid values and calculate summary statistics. After the checks, it will calculate the main correlation and save the results in a JSON file. We will keep the analysis settings in a configuration file so that another person can repeat the same steps.

Before collecting real responses, we will test the whole process on a synthetic sample of 5–10 rows stored in data/sample/. These example rows are invented only to check that the script runs. They will never be described as actual student survey results.

Our GitHub repository will contain README.md, .gitignore, LICENSE, a dependency file with exact versions, Dockerfile, configs/config.json, src/benchmark.py and the synthetic CSV sample. README.md will explain how to run the pipeline in a few commands. The repository should also document the output file and the meaning of its metrics.

8. Ethics and limitations

We will tell students what the questionnaire is about and that participation is optional. We will avoid collecting directly identifying information. Any real response data will be handled carefully and will not be uploaded publicly unless they are appropriately anonymized and sharing is permitted.

There are several limitations. Students may not remember their advertising exposure accurately. Their answers may also be affected by personal preferences or previous shopping experiences. Convenience sampling can introduce selection bias, and a one-time survey cannot establish cause and effect. We will discuss these limitations when interpreting the results.

9. Connection to Task 1 and next steps

This passport continues our Task 1 proposal and uses the same topic, authors and three research questions. Task 1 focused on the literature and research gap. In Task 2, we turn those ideas into measurable variables, clear metrics and a reproducible analysis plan. Our next step is to create and test the GitHub pipeline before collecting actual questionnaire responses in the following stage.